

4.5.17. Prove directly from the definitions that for every integer n , $n^2 - n + 3$ is odd. Use division into two cases: n is even and n is odd.
Suppose n is an integer.

We must show that $n^2 - n + 3$ is odd using cases.

Case 1: n is even

Definition of even:

" n is even $\leftrightarrow n = 2k$, for some $k \in \mathbb{Z}$ "

Let $n, m, k \in \mathbb{Z}$

$$n = 2m$$

$$n^2 - n + 3 = (2m)^2 - (2m) + 3 =$$

$$= 4m^2 - 2m + 3 =$$

$$= 2(2m^2 - m + 1) + 1$$

$$k = 2m^2 - m + 1$$

$$n^2 - n + 3 = 2k + 1$$

$2k + 1$ fulfills the definition of odd,

for some $k \in \mathbb{Z}$.

Therefore, if n is even, then $n^2 - n + 3$

is odd.

Case 2: n is odd

Definition of odd:

" n is odd $\leftrightarrow n = 2k + 1$, for some $k \in \mathbb{Z}$ "

Let $n, m, k \in \mathbb{Z}$

$$n = 2m + 1$$

$$n^2 - n + 3 = (2m + 1)^2 - (2m + 1) + 3 =$$

$$= 4m^2 + 4m + 1 - 2m - 1 + 3 =$$

$$= 4m^2 + 2m + 3$$

$$= 2(2m^2 + m + 1) + 1$$

$$k = 2m^2 + m + 1$$

$$n^2 - n + 3 = 2k + 1$$

$2k + 1$ fulfills the definition of odd,

for some $k \in \mathbb{Z}$.

Therefore, if n is odd, then $n^2 - n + 3$

is odd.

Solution: If n is even or n is odd, then $n^2 - n + 3$ is odd